APT (Automatically Programmed Tool)

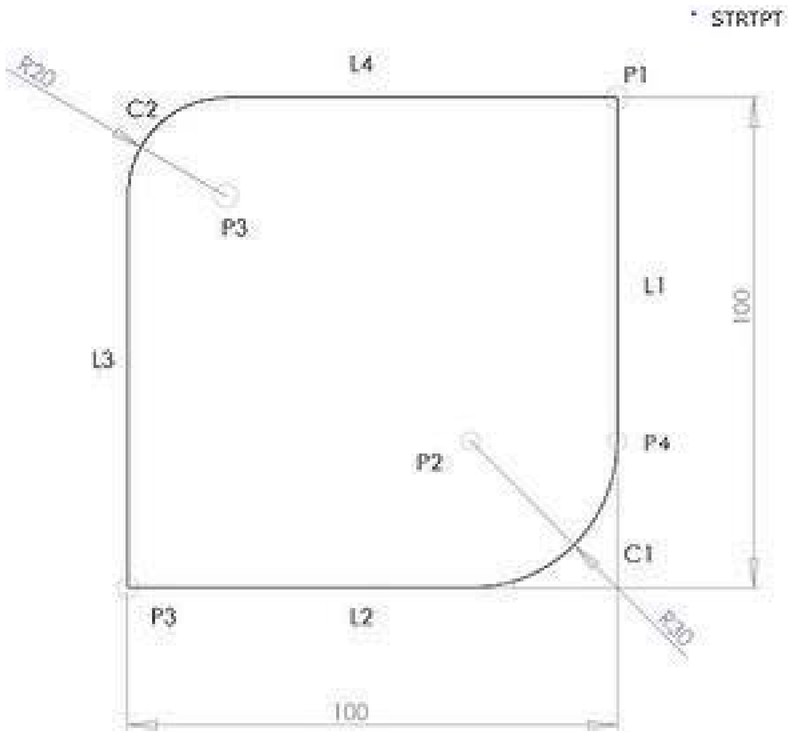
APT (Automatically Programmed Tool) is a high-level computer programming language most commonly used to generate instructions for numerically controlled machine tools.

APT is used to program CNC machine tools to create complex parts using a cutting tool moving in space. It is used to calculate a path that a tool must follow to generate a desired form. APT is a special-purpose language and the predecessor to modern CAM systems. It was created and refined during the late 1950s and early 1960s to simplify the task of calculating geometry points that a tool must traverse in space to cut the complex parts required in the aerospace industry. It was a direct result of the new numerical control technology becoming available at that time and the daunting task that a machinist or engineer faced calculating the movements of the machine for the complex parts for which it was capable. Its development was centered at the same MIT labs that hosted the Numerical Control and the Milling Machine Projects. APT also was Air Force sponsored and is notable for being the world's first major cooperative programming venture, combining government agencies, universities, and a 14-company team organized within the Aircraft Industries Association [now Aerospace Ind Assoc.]. APT was created before graphical interfaces were available, and so it relies on text to specify the geometry and toolpaths needed to machine a part. The original version was created before even FORTRAN was available and was the very first ANSI standard. Later versions were rewritten in FORTRAN. As a U. S. government funded project APT was placed in the public domain.

APT shares many similarities with other computer programming languages such as FORTRAN. A general-purpose computer language takes source text and converts the statements to instructions that can be processed internally by a computer. APT converts source statements into programs for driving numerically-controlled machine tools. The output from an APT processor may be a CL or "Cutter Location" file which is then run through a post-processor specific to the desired control - machine pair. The resulting file is then run by the control of the machine to generate tool motions and other machine actions. Most commonly, this file is in some form of RS-274 format instructions, commonly known as G-code.

Further derivatives of APT were developed, initially mainly to allow the programs to run on minicomputers instead of mainframes. Machine tool makers systems utilize elements of Apt to this day. Standards Developers like STEP-NC took toolpath curves from APT and other sources.

Example Program

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PARTNO / APT-1
CLPRNT
UNITS / MM
NOPOST
CUTTER / 20.0

$$ GEOMETRY DEFINITION
SETPT = POINT / 0.0, 0.0, 0.0
STRTPT = POINT / 70,70,0
P1 = POINT / 50, 50, 0
P2 = POINT / 20, -20, 0
C1 = CIRCLE / CENTER, P2, RADIUS, 30
P3 = POINT / -50, -50, 0
P5 = POINT / -30, 30, 0
C2 = CIRCLE / CENTER, P5, RADIUS, 20
P4 = POINT / 50, -20, 0
L1 = LINE / P1, P4
L2 = LINE / P3, PERPTO, L1
L3 = LINE / P3, PARLEL, L1
L4 = LINE / P1, PERPTO, L1
PLAN1 = PLANE / P1, P2, P3
PLAN2 = PLANE / PARLEL, PLAN1, ZSMALL, 16

$$ MOTION COMMANDS
SPINDL / 3000, CW
FEDRAT / 100, 0
FROM / STRTPT
GO/TO, L1, TO, PLAN2, TO, L4
TLLFT, GOFWD / L1, TANTO, C1
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GOFWD / C1, TANTO, L2
GOFWD / L2, PAST, L3
GORGT / L3, TANTO, C2
GOFWD / C2, TANTO, L4
GOFWD / L4, PAST, L1
NOPS
GOTO / STRTPT
FINI
```

CNC Machine Language - G-Code List

G-Code is one of a number of computer code languages that are used to instruct CNC machining devices what motions they need to perform such as work coordinates, canned cycles, and multiple repetitive cycles. Industry has standardized on G-Code as its basic set of CNC machine codes.

G-Code is the most popular programming language used for programming CNC machinery. Some G words alter the state of the machine so that it changes from cutting straight lines to cutting arcs. Other G words cause the interpretation of numbers as millimeters rather than inches. Some G words set or remove tool length or diameter offsets.

Below is a complete listing of current codes.

G-Code	Description
G00	Rapid Linear Interpolation
G01	Linear Interpolation
G02	Clockwise Circular Interpolation
G03	Counter Clockwise Circular Interpolation
G04	Dwell
G05	High Speed Machining Mode
G10	Offset Input By Program
G12	Clockwise Circle With Entrance And Exit Arcs
G13	Counter Clockwise Circle With Entrance And Exit Arcs
G17	X-Y Plane Selection
G18	Z-X Plane Selection
G19	Y-Z Plane Selection
G28	Return To Reference Point
G34	Special Fixed Cycle (Bolt Hole Circle)
G35	Special Fixed Cycle (Line At Angle)
G36	Special Fixed Cycle (Arc)

G37	Special Fixed Cycle (Grid)
G40	Tool Radius Compensation Cancel
G41	Tool Radius Compensation Left
G42	Tool Radius Compensation Right
G43	Tool Length Compensation
G44	Tool Length Compensation Cancel
G45	Tool Offset Increase
G46	Tool Offset Decrease
G50.1	Programmed Mirror Image Cancel
G51.1	Programmed Mirror Image On
G52	Local Coordinate Setting
G54 - G59	Work Coordinate Registers 1 Thru 6
G60	Unidirectional Positioning
G61	Exact Stop Check Mode
G65	Macro Call (Non Modal)
G66	Macro Call (Modal)
G68	Programmed Coordinate Rotation
G69	Coordinate Rotation Cancel
G73	Fixed Cycle (Step)
G74	Fixed Cycle (Reverse Tapping)
G76	Fixed Cycle (Fine Boring)
G80	Fixed Cycle Cancel
G81	Fixed Cycle (Drilling / Spot Drilling)
G82	Fixed Cycle (Drilling / Counter Boring)
G83	Fixed Cycle (Deep Hole Drilling)
G84	Fixed Cycle (Tapping)
G85	Fixed Cycle (Boring)
G86	Fixed Cycle (Boring)
G87	Fixed Cycle (Back Boring)
G88	Fixed Cycle (Boring)
G89	Fixed Cycle (Boring)
G90	Absolute Value Command
G91	Incremental Value Command
G92	Work Offset Set

G101	User macro 1 (substitution) =
G102	User macro 1 (addition) +
G103	User macro 1 (subtraction) -
G104	User macro 1 (multiplication) *
G105	User macro 1 (division) /
G106	User macro 1 (square root)
G107	User macro 1 (sine) sin
G108	User macro 1 (cosine) cos
G109	User macro 1 (arc tangent) tan
G110	User macro (square root)
G200	User macro 1 (unconditional branch)
G201	User macro 1 (zero condition branch)
G202	User macro (negative condition branch)